

Esp32Tap Breadboard — Logical Cluster Layout

Approximate placement only. Use two full-size breadboards side-by-side; keep each zone compact, then connect zones by the named nets below.

Start unpowered. Leave STANDALONE POWER and COIL POWER open. Meter-identify relay, transistor, regulator, and breakout pins before inserting them.

Breadboard A — treadmill / power side

+8V_RAW: short rail, connector end only

1 — RJ45 + relay contacts

- CONSOLE and MOTOR breakouts
- K1 at right edge of zone
- pins 1/7 → GND
- pins 2/8 → +8V_RAW
- pins 3/4/5 pass through
- pin 6 through K1 contacts

Put RJ45 breakouts at the outer edge.

+8V_RAW

2 — Input power

- RXEF075 + 1N5822
- P6KE12A TVS
- TSR1-2433E
- 22 μ F + two 10 μ F caps
- local VIN and TSR_3V3 strips

Keep protection parts near +8V entry.

VIN

4 — Voltage monitor

- TPS3700 breakout
- UV divider: 150k / 10k
- OV divider: 255k / 10k
- two 1 nF filters
- TREAD_OK pulls + GPIO6 R

Keep divider midpoints short.

K1 coil

5 — Relay driver

- TPS709 + 1 μ F / 4.7 μ F
- BC337 + 560 Ω + 10k
- P6KE6.8CA across K1 coil
- COIL POWER removable jumper

Place beside K1; verify B/C/E leads.

GND: join both boards with two jumpers

Breadboard B — ESP32 / logic side

LOGIC_3V3: logic board only

3 — ESP32 DevKit

Straddle center ditch; USB faces edge.

Left-facing jumpers:

- GPIO6 — TREAD_OK
- GPIO21 — RELAY_CMD
- GPIO15 — TX_ENABLE
- GPIO17 — UART TX
- GPIO18 / 16 — RX taps

Right-facing jumpers:

- GPIO4 / 5 — K1 feedback
- GPIO7 — VBUS_PRESENT_N
- GPIO38 — status LED

Rails:

- 3V3 → LOGIC_3V3
- GND → common GND
- 5V/VBUS → zone 7 only

Leave room for USB cable and reset.

GPIOs

6 — Logic + UART

- SN74AHC08N across ditch
- SN74AHC126N across ditch
- 100 nF at each IC
- command pull-downs
- R_TX 100 Ω
- two 10k RX resistors

IC notches point the same way.

Tie every unused input low.

7 — Feedback + LEDs

- K1 feedback pull-ups
- 2N7000 VBUS sense
- green status LED + 330 Ω
- red power LED + 1k
- C_LOGIC 100 nF

GPIO4/5/7/38

INTER-BOARD JUMPERS: SEE PAGE 2

Cluster Build Order and Inter-Zone Jumpers

Build and meter-check one zone at a time. Use the point-to-point wiring PDF for every pin-level connection.

Suggested build order

Order	Zone	Stop-and-check point
1	3 — ESP32	USB boot works; 3V3 and GND rails are correct.
2	7 — Feedback/LEDs	Power LED works; GPIO pins are not shorted.
3	6 — Logic/UART	DIP orientation, VCC/GND, unused inputs, and decouplers checked.
4	1 — RJ45/K1	Pass-through continuity correct with K1 unpowered.
5	2 — Input power	Polarity checked; VIN and GND have no short. Do not join standalone power yet.
6	4 — Monitor	Divider values and TREAD_OK pull resistors meter correctly.
7	5 — Relay driver	BC337 lead order and coil TVS checked; leave COIL POWER open.

Rail policy

Rail/net	Where	Rule
GND	Both boards	Join boards with two short jumpers; all USB and treadmill grounds are common.
+8V_RAW	Zone 1 → zone 2 only	Do not run along the logic board.
VIN	Zones 2, 4, 5	Keep on treadmill/power board.
LOGIC_3V3	Zones 3, 4, 6, 7	Powered from DevKit USB during initial bring-up.
TSR_3V3	Zone 2	Join to LOGIC_3V3 only through removable STANDALONE POWER jumper.
+5V_RLY	Zone 5	Short local strip only; coil connection stays removable.

Major inter-zone jumpers

Zones	Named nets / purpose
1 → 2	+8V_RAW, GND — protected local-power input.
2 → 4	VIN, GND — monitor supply and divider source.
2 ≠ 3	STANDALONE POWER — TSR_3V3 to LOGIC_3V3; install only when USB is absent.
3 ≠ 4	LOGIC_3V3, TREAD_OK_MCU/GPIO6.
4 → 6	TREAD_OK — hardware permission to both AHC08 gates.
3 → 6	GPIO21 RELAY_CMD, GPIO15 TX_ENABLE, GPIO17 UART TX.
6 → 5	RELAY_GATE — gated command to TPS709 enable and BC337 base resistor.
5 ≠ 1	K1 coil pins 1 and 16 — keep short; COIL POWER remains removable.
6 → 1	R_TX output to K1 pin 8 — switched ESP transmit.
1 → 3	CONSOLE pin 6 via 10k → GPIO18; pin-3 net via 10k → GPIO16.
1 → 7	K1 pins 9, 11, 13 — NO/COM/NC feedback contacts.
7 ≠ 3	GPIO4, GPIO5, GPIO7, GPIO38, 5V/VBUS, LOGIC_3V3.

Fast physical rules

- Put every DIP IC across a breadboard center ditch, with notches pointing the same direction.
- Put each 100 nF capacitor directly beside its IC supply pins.
- Keep UV_SENSE and OV_SENSE resistor/capacitor nodes inside zone 4.
- Keep BC337, its two resistors, coil TVS, and K1 coil pins together.
- Use colored jumpers consistently: black GND, red raw/VIN, blue 3V3, yellow control, green feedback.
- Label both ends of every jumper that crosses between boards.

This page groups the circuit; it does not replace the 126-row From/To wiring list. Never infer a physical pin from package shape—meter or datasheet-verify it.